

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

C05: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications. (BTL 5)*

Assessment Tool Weightage: 50%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Reverse Engineering

Code of Conduct:

I K.ASWINI(192312397)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

K.ASWINI

Signature of the student:

Reverse Engineering

1. Machine Learning Techniques for Reverse Engineering

Description:

Machine Learning can be used to classify software components, identify programming patterns, detect anomalies, and predict the functionality of unknown code. Students study supervised, unsupervised, and semi-supervised learning approaches used in reverse engineering.

Student Activity:

- Collect a dataset of source-code samples.
- Extract code features.
- Apply classification or clustering algorithms.
- Analyze similarities among programs.
- Compare the performance of different ML algorithms.

Expected Outcome:

Students will evaluate the suitability of different ML algorithms for software reverse-engineering tasks.

2. AI for Source Code Analysis and Program Understanding

Description:

AI can analyze large source-code repositories and automatically identify functions, classes, variables, dependencies, and programming patterns. Natural Language Processing and Large Language Models can also convert complex code into human-readable explanations.

Student Activity:

- Select a source-code repository.
- Analyze the code using AI-assisted tools.
- Identify important functions and dependencies.
- Generate natural-language explanations.
- Verify the AI-generated explanations against the actual code.

Expected Outcome:

Students assess the effectiveness and limitations of AI in understanding unfamiliar source code.

3. AI-Based Malware Reverse Engineering and Analysis

Description:

AI can assist security researchers in analyzing malware behavior by identifying suspicious code patterns, API calls, strings, functions, and execution behaviors. Students can study the conceptual workflow of AI-assisted malware analysis using **safe, non-executable datasets or publicly available research samples**.

Student Activity:

- Study malware-analysis datasets.
- Identify relevant static features.
- Apply ML classification techniques.
- Analyze suspicious patterns.
- Evaluate classification accuracy.

Expected Outcome:

Students understand and evaluate how AI can support automated malware analysis.

4. Reverse Engineering of APIs Using AI

Description:

APIs may need to be understood when documentation is incomplete or unavailable. AI can analyze API requests, responses, source code, and observed behavior to infer endpoints, parameters, data formats, and relationships.

Student Activity:

- Select a documented/open API or controlled test API.
- Analyze its request-response structure.
- Use AI to infer API functionality.
- Generate documentation.
- Compare generated documentation with the actual API specification.

Expected Outcome:

Students assess the usefulness of AI for API understanding and documentation recovery.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood

Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

ANSWERS

1. Machine Learning Techniques for Reverse Engineering

Explanation

Machine Learning helps in reverse engineering by analyzing source code and identifying patterns automatically. It can classify programs, detect similarities, and even predict what unknown code does.

There are three main approaches:

- **Supervised Learning:** Uses labeled data to classify code (e.g., malware vs normal)
- **Unsupervised Learning:** Groups similar programs without labels (clustering)
- **Semi-supervised Learning:** Uses a mix of labeled and unlabeled data

Student Activity (Explanation)

Students first collect different source code samples from repositories. Then they extract features such as keywords, function calls, and structure. After that, they apply machine learning algorithms like Decision Trees or K-Means to classify or cluster the code. Finally, they compare results of different algorithms.

Expected Outcome

Students understand which ML algorithms work best for analyzing and classifying software during reverse engineering.

2. AI for Source Code Analysis and Program Understanding

Explanation

AI can read and understand source code similar to how humans do. It identifies functions, variables, classes, and relationships between them. Advanced AI models like NLP and LLMs can even convert code into simple explanations.

Student Activity (Explanation)

Students choose a code repository and analyze it using AI tools. They identify important components such as functions and dependencies. Then they generate explanations of the code and compare them with actual logic to check accuracy.

Expected Outcome

Students learn how effective AI is in understanding unknown code and where it may fail or give incorrect interpretations.

3. AI-Based Malware Reverse Engineering and Analysis

Explanation

AI helps in analyzing malware safely by studying patterns instead of executing harmful code. It identifies suspicious behavior such as unusual API calls, strings, or code structures.

Student Activity (Explanation)

Students work with safe malware datasets. They extract features like API calls or opcode patterns and apply classification algorithms to detect whether a file is malicious or not. They then evaluate how accurate the model is.

Expected Outcome

Students understand how AI can assist in detecting and analyzing malware without actually running it, improving cybersecurity analysis.

4. Reverse Engineering of APIs Using AI

Explanation

Sometimes APIs do not have proper documentation. AI can analyze API requests and responses to understand how the API works, including endpoints, parameters, and data formats.

Student Activity (Explanation)

Students select an API and study its request-response patterns. Using AI tools, they try to infer how the API functions and automatically generate documentation. Then they compare it with the original documentation.

Expected Outcome

Students learn how AI can help in understanding and reconstructing API documentation when it is missing or incomplete.